Naive Bayes Classifier

i) Start

ii) Input: · Training dataset with features X
· Class labels Y

iii) Compute prior probability for each class:

$$P(C_k) \leftarrow \frac{\text{No. of samples in class } C_k}{\text{Total no. of samples}}$$

iv) For a new set sample:

a) for each class C_k, compute posterior probability using Bayes Theorem: $P(C_k|X) \leftarrow P(C_k) \prod P(x_i|C_k)$

b) Ignore denominator since it is constant for all classes.

v) For each feature and each class:
· Compute likelihood ($P(x_i|C_k)$) (for continuous data, use Gaussian formula)

vi) Assign the class with the highest posterior probability.

vii) Output predicted class.

viii) End.